Worksheet 47 · Functions IGCSE

Edexcel IGCSE Mathematics · past paper questions

Name:	Subject: Math	Date:
Learning Objectives: Students will be able to: • Evaluate functions using function notation • Find composite functions • Find inverse functions and state domain restrictions		
Grade: / 98	**Teacher's Feedback:**	

Answer in the space given. Show your working. Marks for each part are shown in brackets.

Key notes

Domain → Set of all inputs, x Range → Set of all outputs, y

Excluding values from domain → Can't: $\sqrt{\text{negative}}$ or divide by 0.

Composite Functions → $fg(x)$ Means do g then f.

Inverse Functions → $y = f(x) \leftrightarrow x = f^{-1}(y)$

Questions *Functions*

Question 1 (4 marks)

The function f is such that $f(x) = (x - 4)^2$ for all values of x.

(a) Find f(1)

(1)

(b) State the range of the function f.

(1)

The function g is such that $g(x) = \frac{4}{x+3}$ $x \neq -3$

(c) Work out fg(2)

(2)

Question 2 (32 marks)

The function f is such that $f(x) = x^2 - 8x + 5$ where $x \leqslant 4$

Express the inverse function f^{-1} in the form $f^{-1}(x) = ...$

$f^{-1}(x) =$ .....

Question 3 (7 marks)

The function f is such that

$$f(x) = \frac{3x - 5}{4}$$

(a) Find f(−7)

(1)

(b) Express the inverse function f^{-1} in the form $f^{-1}(x) = ...$

$f^{-1}(x) =$ .....

(2)

The function g is such that

$$g(x) = \sqrt{19 - x}$$

(c) Find fg(3)

(2)

(d) Which values of x cannot be included in any domain of g?

(2)

Question 4 (6 marks)

The functions f and g are defined as

$$f(x) = \frac{x}{4x-3} \quad \text{and} \quad g(x) = x - 5$$

(a) State which value of x must be excluded from any domain of the function f.

.....
(1)

(b) Find $fg(x)$.
Simplify your answer.

$fg(x) =$ .....
(2)

(c) Express the inverse function f^{-1} in the form $f^{-1}(x) = \ldots$

$f^{-1}(x) =$ .....
(3)

Question 5 (5 marks)

The functions f and g are such that

$$f(x) = x^2 - 2x \qquad g(x) = x + 3$$

The function h is such that $h(x) = fg(x)$ for $x \geqslant -2$

Express the inverse function $h^{-1}(x)$ in the form $h^{-1}(x) = \ldots$

$h^{-1}(x) =$

Question 6 (6 marks)

The function f is defined as $f(x) = \frac{\sqrt{x^2 + k^2}}{x}$ for $x > 0$ and where k is a positive number.

a) Find the value of p for which $f^{-1}(p) = k$

$p =$ .....
(3)

The function g is defined as $g(x) = x^2$ for $x > 0$

b) Given that $gf(a) = k$ for $k > 1$
find an expression for a in terms of k.

$a =$ .....
(3)

Question 7 (4 marks)

The function f is defined as

$$f: x \mapsto \frac{2x}{x-6} \qquad x \neq 6$$

(a) Find f(10)

.....
(1)

(b) Express the inverse function f^{-1} in the form $f^{-1}: x \mapsto \ldots$

$f^{-1}: x \mapsto$ .....
(3)

Question 8 (4 marks)

The function f is such that $f(x) = \frac{3}{x-2}$

(a) Find f(1)

.....
(1)

(b) State which value of x must be excluded from any domain of f

.....
(1)

The function g is such that $g(x) = x + 4$

(c) Calculate fg(2)

.....
(2)

Question 9 (6 marks)

The functions f and g are such that

$$f(x) = 2x - 3$$
$$g(x) = \frac{x}{3x+1}$$

(a) State the value of x that cannot be included in any domain of g

.....
(1)

(b) Find $gf(x)$
Simplify your answer.

$gf(x) =$ .....
(2)

(c) Express the inverse function g^{-1} in the form $g^{-1}(x) = \ldots$

$g^{-1}(x) =$ .....
(3)

Question 10 (9 marks)

The functions f and g are defined as

$f(x) = x^2 + 6$
$g(x) = x - 10$

(a) Find fg(3)

.....
(2)

(b) Solve the equation $fg(x) = f(x)$
Show clear algebraic working.

.....
(3)

The function h is defined as

$$h(x) = \frac{2x-4}{x}$$

(c) State the value of x that cannot be included in the domain of h

.....
(1)

(d) Express the inverse function h^{-1} in the form $h^{-1}(x) = \ldots$

$h^{-1}(x) =$ .....
(3)

Question 11 (6 marks)

The functions f and g are defined as

$f(x) = 5x^2 - 10x + 7$ where $x \geqslant 1$

$g(x) = 7x - 6$

(a) Find fg(2)

.....

(2)

(b) Express the inverse function f^{-1} in the form $f^{-1}(x) = ...$

$f^{-1}(x) =$ .....

(4)

Question 12 (5 marks)

The function g is defined as

$g: x \mapsto 5 + 6x - x^2$ with domain $\{x : x \geqslant 3\}$

(a) Express the inverse function g^{-1} in the form $g^{-1}: x \mapsto ...$

$g^{-1}: x \mapsto$ .....

(4)

(b) State the domain of g^{-1}

.....

(1)

Question 13 (4 marks)

The functions f and g are such that

$f(x) = x + 25$ $g(x) = x^2 - 12x$

The function h is such that $h(x) = fg(x)$

The domain of h is $\{x : x \leqslant 6\}$

Express the inverse function h^{-1} in the form $h^{-1}(x) = ...$

$h^{-1}(x) =$ .....